# Childhood Cancer: Outcomes, Biology, Care, and Survivorship

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**Scope:** Narrative synthesis of National Cancer Institute (NCI) childhood and adolescent cancer information and American Cancer Society (ACS) resources on childhood leukemia survival, cancer genetics, biomarkers, chemical carcinogens, and the scope of ACS early-detection guidelines. This is an educational summary, not a systematic review or clinical guideline.


## Abstract

Childhood cancer is uncommon, encompasses biologically distinct diseases, and should not be treated as one prognosis. U.S. estimates published by NCI project 14,910 diagnoses and 1,590 deaths among people aged 0–19 in 2024 [1]. NCI reports age-group-specific 5-year survival estimates from 2013–2019 ranging from 83.2% to 87.8% [1]. For childhood leukemia, ACS gives approximate overall 5-year survival figures of about 90% for acute lymphoblastic leukemia (ALL), 65–70% for acute myeloid leukemia (AML), and more than 92% for chronic myeloid leukemia (CML); outcomes vary with disease and clinical factors [2]. NCI estimates inherited cancer-predisposition variants cause about 8–10% of childhood cancers overall, with substantial variation by cancer type [1]. These population estimates do not predict an individual child's outcome. [1, 2]


## 1. Context and scope

"Childhood cancer" includes multiple diseases rather than a single condition. NCI identifies leukemia, malignant brain and other central nervous system tumors, lymphomas, and several other solid tumors among the common groups; the distribution differs by age [1]. Statistics therefore need a defined age group, cancer type, diagnosis period, and outcome measure before comparison [1, 2].

The figures in this brief describe U.S. populations and different observation periods. They are not forecasts for a particular child and should not be combined as if they came from one cohort [1, 2].

## 2. Incidence and outcomes

NCI estimates that 14,910 people ages 0–19 would be diagnosed with cancer and 1,590 would die of cancer in the United States in 2024 [1]. These are estimates, not a count of cases from a completed 2024 registry year [1].

NCI reports 5-year survival by age for diagnoses during 2013–2019: 83.2% for children younger than 1 year, 87.8% for ages 1–4, 85.7% for ages 5–9, 85.5% for ages 10–14, and 87.3% for ages 15–19 [1]. These aggregate age-band outcomes conceal important differences among tumor types and subgroups; NCI notes that outcomes have improved over decades but remain poor for some cancers [1].

A 5-year survival rate is a population statistic. ACS explains that childhood leukemia survival figures are based on children diagnosed and treated in earlier years; improvements in treatment may change outcomes for children diagnosed today [2]. The treating team is best placed to explain how age, subtype, risk classification, genetics, response, and other clinical details affect an individual case [2].

## 3. Leukemia illustrates disease-specific variation

Leukemia is not a single disease. ACS describes approximate overall 5-year survival rates of about 90% for childhood ALL, 65–70% for AML, and more than 92% for CML [2]. These are broad estimates. For ALL, prognostic factors include age at diagnosis, initial white blood cell count, leukemia-cell genetic features, and response to treatment; AML outlook varies with subtype, genetic risk group, and early response [2]. CML is rare in children, and some patients may remain on treatment for many years [2].

These figures should not be ranked or applied to an individual without clinical context. ACS describes survival rates as estimates and encourages families to discuss their child's situation with the care team [2].

## 4. Genetics and inherited risk

Cancer involves genetic changes, but a genetic change is not necessarily inherited from a parent. NCI estimates that inherited pathogenic variants in cancer-predisposition genes cause about 8–10% of childhood cancers overall; the proportion varies substantially by cancer type, and inherited variants are more common in some diagnoses such as retinoblastoma [1]. ACS explains that many cancer-associated genetic changes arise during a person's lifetime rather than being inherited [3].

A child's diagnosis does not automatically mean that relatives carry an inherited risk [1, 3]. When personal or family history suggests a hereditary syndrome, families can ask clinicians whether evaluation by a genetics professional is appropriate; testing decisions and interpretation belong with qualified clinicians and genetic counselors [3].

## 5. Chemical exposures and childhood cancer evidence

ACS provides general information about chemicals and cancer, including resources on known and probable human carcinogens [5]. That general information should not be interpreted as evidence that a particular exposure caused an individual child's cancer. NCI specifically notes that environmental causes of childhood cancer have been difficult to identify [1]. This brief therefore makes no child-specific causal claim about chemical exposure; concerns about a suspected exposure should be discussed with qualified health professionals and public-health authorities.

## 6. Biomarkers and treatment discussions

Biomarkers are genes, proteins, or other substances that can be tested to learn more about a cancer. ACS notes that biomarker test results may help clinicians consider treatment options, assess how likely a cancer is to grow or spread, evaluate whether treatment is working, or identify recurrence [4]. Whether testing is useful depends on the cancer and the clinical situation; families can ask the care team what a specific test could and could not tell them [4].

## 7. Early detection and screening scope

ACS describes an evidence-review and guideline-development program for cancer screening, with listed guidance for colorectal, cervical, and lung cancer, among others [6]. These materials are not a childhood-cancer screening guideline. This page should therefore be used to understand ACS's general early-detection work, not as a recommendation to screen an asymptomatic child [6]. Any concern about a child's symptoms or risk should be discussed with the child's clinician.

## 8. Care setting, trials, and family support

NCI notes that children and adolescents are often treated at children's cancer centers with expertise in pediatric cancers and comprehensive family care [1]. Clinical trials have contributed to treatment advances, but participation is voluntary; families can discuss whether a trial is appropriate with the treatment team [1]. Families can also ask about diagnosis and subtype, treatment options, expected benefits and risks, supportive care, and whom to contact with concerns [1].

## 9. Survivorship and long-term follow-up

Completing treatment does not always end cancer-related care. NCI describes late effects that can arise months or years afterward; their likelihood depends on the cancer, treatment exposures, and age at diagnosis [1]. Regular follow-up can help identify and manage health issues. NCI recommends retaining a treatment summary, including diagnosis, therapies and doses, radiation fields, surgeries, complications, genetic testing information, and treatment completion date, to support care across the transition into adulthood [1].

## 10. Conclusions

The evidence supports four practical conclusions: childhood cancer is a collection of distinct diseases; population survival rates require careful interpretation; inherited predisposition explains a minority of childhood cancers overall but is important in selected diagnoses and families; and care planning includes specialist treatment, biomarker discussions where relevant, voluntary research options, and survivorship follow-up [1–4]. Families should use population information to prepare questions, not to predict an individual child's outcome.

## Sources


1. National Cancer Institute. Cancer in Children and Adolescents. Updated August 27, 2024. U.S. incidence estimates refer to 2024; age-specific survival figures refer to 2013–2019 unless otherwise stated. Locally ingested text.
2. American Cancer Society. Prognostic Factors and Survival Rates for Childhood Leukemia. Last revised July 22, 2025. Locally ingested text.
3. American Cancer Society. Genetics and Cancer Risk. Accessed October 5, 2026. Locally ingested text.
4. American Cancer Society. Cancer Biomarkers. Patient fact sheet No. 080826, Rev. 8/25. Locally downloaded PDF.
5. American Cancer Society. Chemicals and Cancer. Accessed October 5, 2026. Locally ingested text.
6. American Cancer Society. Early Cancer Detection Science. Accessed October 5, 2026. Locally ingested text.


## Limitations

This brief summarizes a small set of U.S. public-facing sources and does not independently reanalyze registry data or appraise every cited study. The NCI page was updated in 2024, leukemia material was last revised in 2025, and source data periods differ. Statistics may be updated after this brief's preparation date. This document does not provide diagnosis, treatment recommendations, or a personal prognosis.